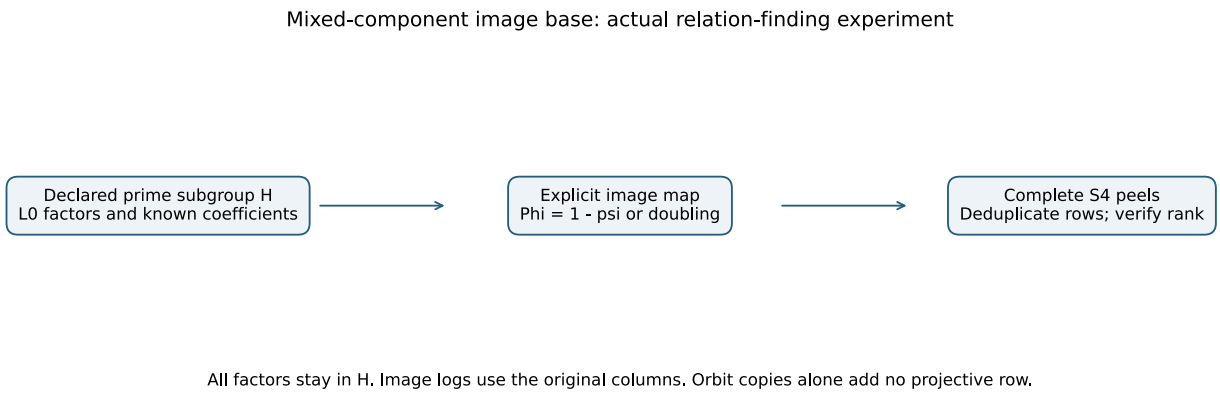
Actual execution boundary and inference limits



The measured object is the frozen v1 native program, source archive SHA-256 61948c5aaae1e1451dee1760e619ffc4d585c11808c90a8cff60dce2bd5786c1. Its source is not replaced when the report or resume machinery changes.

Inputs and the cold clock

The clock in native/src/cold.rs::execute starts after loading and checking the frozen tape, and before build and the exact symbolic S4 derivation. It stops after back substitution and group verification of the recovered target and every usable factor. Evidence serialization follows the clock. All failed peels and complete queries up to rank or the original cap are inside it. Input parsing, process startup and known-answer fixture creation are outside it. The public curve/subgroup/eigenvalue parameters are given; point counting and discovery of those domain parameters are not timed.

Each comparison holds the supplied twisted curve and subgroup fixed and changes the factor-base image. It isolates the proposed phi-image mechanism against doubling on that model. It does not compare replacing a curve by a different twist, or assign the model's order-three automorphism exclusively to twisting. The explicit equation and field in MATHEMATICS.md define the experiment; “quartic” here describes the extension-field construction.

The legacy tape contains known answers and reference diagnostics. The candidate code reads the public G, Q and only entries 0 and 1 of each query record; it does not read known_scalar, reference logs, hits or ranks to construct relations or decide when to stop. This is a source-audited known-answer experiment, not a process-isolated secret-input experiment.

The runner also reads base_coordinates for public physical IDs, counts and sign orientation, and checks each point against the independently computed lift. It enumerates every required abscissa and performs its Tonelli--Shanks lift inside the clock; it never substitutes an encoded y coordinate for that lift. Orbit columns, coefficients, required images, action checks, exceptional handling and zero-triple tables are rebuilt. Active arms construct only their required L0 factors and images.

Clarification of the frozen supplement: the phrase “no precomputed lift … is available” in SUPPLEMENT_COLD_COST.md is too strong as a data isolation description: the public coordinate encoding is present for comparison and sign alignment. The computed lifts are charged in full. Report the actual supplied encoding and exclusions above; do not describe this program as accepting only G,Q with all other fixture information physically absent. This clarification does not edit the frozen protocol, raw source, timings, output sets or acceptance thresholds.

Complete elapsed intervals versus incomplete operation prices

Each cold record contains setup and five exclusive online phases. The verifier requires their exact sums to match online and cold nanoseconds. All computation of this instrumented implementation, including allocation, normalization, hash/tree work, roots and evidence bookkeeping, is charged to those elapsed intervals. Allocation telemetry adds overhead and is included on every arm. Resident memory and live Rust allocation peaks are different measures and are reported separately.

The field/group meter and F4 row-reduction counter have narrower scopes. They are not a complete GAE operation price. In particular, not every F4 normalization/root/hash operation emits a separate native-operation count. The required 0/0.1/1 GAE band, matched strong-rho measurement and validation gate therefore remain unfulfilled. No leaderboard ranking, rho advantage or complete operation-cost speedup follows from the elapsed-time ratios.

The external control measures complete reduced Gröbner bases on the same 378 exported inputs. It excludes polynomial restriction, roots, group work, matrix work and the rest of the IC pipeline. Its internal timers exclude process startup. It does not price an msolve-based end-to-end DLP solver. Positions use query zero and the frozen first/middle/last peels; the sample is not a benchmark of the whole solver's difficult tail. No pipeline cost is projected from its component ratios. Magma, FGb, Singular and OpenF4 comparisons are not reproduced in this round; no claim to the fastest external implementation is made.

Timing admission after interruption

The cold resource envelope is c7i.2xlarge, one reserved whole SMT core, one Rayon thread, an 8 GiB process address-space limit and 1,200 seconds per arm. Every measured record must have exit 0, uncontended isolation, no other user thread left on the reserved core, the declared CPU topology and the recorded address-space/watchdog command. A resource refusal is retained.

A resumed timed unit contains all ten arms and the additional original A/A execution from one originating job. Only complete admitted units are reused. An incomplete or contended round is restarted in full; its earlier records remain. Selection is chronological, never by elapsed time. Final ratios pair reference and candidate within each originating round. Raw medians across several hosts of the same class are descriptive.

The inactive-column phi-L0 arm is censored on all 21 inputs. Its elapsed cost is consumed work at the stream cap, not cost to a verified solution. It cannot appear as a successful cheap arm in a ratio.

Population and claims

The panel comprises seven fields and three seeded records per field, all with public scalar 271. Historical stopping of the frozen streams can bias empirical relation rates. Timing repeats use the same target and trajectory. They are not independent draws of secret scalars, independent curve families, or a basis for an asymptotic claim. Every per-target row and every failure is retained. Aggregate tuple and operation counts are diagnostics, not a multi-target amortization of DLP setup or recovery.